
AutoSafeHarness: Automatically Securing Agents with Model- and Domain-Specific Safety Harnesses

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Abstract

An agent that acts in the real world can be subverted when an attacker either smuggles instructions into the content it reads or sends a malicious request directly as the user instruction. Current defenses are hand-designed once and applied uniformly, yet threat surfaces vary sharply: a detector tuned to one model misfires on another, and a tool gate calibrated for one domain over-blocks in the next. To this end, we present AutoSafeHarness, which searches the scaffolding around a frozen model for a safety harness optimized per model and per domain. AutoSafeHarness treats the harness as a dual-surface artifact (natural-language prompt and pipeline code) and discovers defenses through a closed loop: a Designer proposes candidates from a frontier of prior security designs, a fresh-context Criticizer screens proposals for benchmark overfit by keying acceptance on attack invariants, and a staged cascade evaluates on the target model–domain pair. On AgentDojo, searched harnesses drive attack success to zero while preserving near-baseline utility. On the DecodingTrust-Agent platform, across fifteen model $\times$ domain cells spanning five frontier agents and both attack types, the average attack success rate falls from 45.6% to 10.0% at a benign-utility cost of roughly three points. The fifteen returned harnesses are demonstrably specific: mechanisms shared across cells that share a domain overlap three times more than those shared across cells that share a model.

 **Code:** github.com/SaFo-Lab/HarneSS

 **Website:** andylinx.github.io/Harness_Secure_Website

1 Introduction

An agent that reads the web, opens an email, or pulls a file from a repository can act on what it finds: send money, share documents, edit a live system. That reach is what makes the agent useful, and it is also what makes it vulnerable, from two directions. Anyone who controls the content an agent reads can hide instructions inside it, and the model may follow them as faithfully as if its own user had issued them; this is *indirect prompt injection* [Greshake et al., 2023, Liu et al., 2024, Perez and Ribeiro, 2022]. The request itself can also be adversarial, asking the agent outright to take a harmful or unauthorized action; this is a *direct attack* [Chen et al., 2026]. Neither is a corner case. Both are the structural price of capability: the wider an agent’s reach, the larger the gap between the task its principal intended and the actions an adversary can induce, whether the adversary writes the content the agent reads or issues the instruction itself. As agents are deployed in increasingly consequential settings, that gap becomes a security-critical liability.

A growing line of system-level defenses places the security decision outside the model, in the scaffolding that surrounds it [Xiang et al., 2026]. Prompt-level marking such as spotlighting brackets untrusted content so the model discounts it [Hines et al., 2024]; learned detectors flag injected text [Inan et al., 2023]; plan-constrained monitors such as DRIFT police deviations from an expected tool trajectory [Li et al., 2026a]; and capability interpreters such as CaMeL route untrusted data away

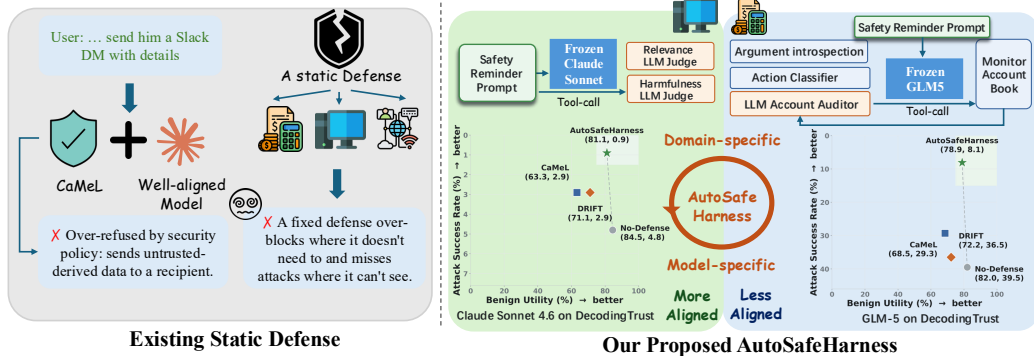


Figure 1: **A searched harness adapts where fixed defenses miscalibrate.** *Left:* CaMeL blocks an in-scope response on Sonnet 4.6/os-filesystem because it treats data derived from untrusted content as exfiltration. *Right:* AutoSafeHarness searches a harness for each model and domain, selecting a light design for Sonnet 4.6 and a deeper design for GLM-5. In both cases, it improves the utility–ASR trade-off over the fixed baselines (DTAP; Table 3).

from control flow and provably block exfiltration [Debenedetti et al., 2025]. These defenses share a common architecture: each is a fixed artifact, designed once by experts, and applied uniformly across models and domains. Yet the threat is not uniform. A detector tuned to one model’s failure modes misfires on another; a monitor calibrated for one toolset over-blocks in the next; a policy framing that a cautious model respects, a more compliant model walks straight through. The artifact is static, the deployment is not, and the mismatch is where attacks live.

We begin from a different premise: the unit worth optimizing is not the model but the harness around it. The model can remain frozen. The ordered pipeline that wraps it (system messages, tool-call gates, execution wrappers, output audits) is entirely ours to write. It is the one place a security decision can sit beyond the attacker’s reach, and, unlike the model, it is programmable and searchable. Two recent results show how far this idea can be pushed for capability: Meta-Harness uses an agentic proposer to rewrite a harness’s source code and improve an application end-to-end [Lee et al., 2026], and Natural-Language Agent Harnesses (NLAH) represent the harness as an editable natural-language policy executed by a runtime [Pan et al., 2026]. We redirect the optimization target toward the safety–utility trade-off, seeking defenses tailored to the exact model and domain they protect.

Concretely, we present AutoSafeHarness, which treats the deployable unit as the pair (M, H) and searches the harness H around a frozen model M until the pair behaves as a secure agent (Figure 2). AutoSafeHarness is a closed, experience-driven loop with four roles. An agentic Designer proposes dual-surface harnesses (a natural-language policy together with gated tool-handling code) warm-started from mature security systems; a fresh-context Criticizer screens every proposal for overfitting, keying acceptance on attack invariants (provenance, scope, intent) rather than the surface vocabulary of any one benchmark; a sandboxed cascade scores benign utility against attack success; and an Analyzer distills the failed traces into experience that steers the next proposal. Three properties make this attractive. It is general: the loop harnesses any model that exposes a tool-use interface. It is specific: because the search reads M ’s own failures, the harness it returns is fitted to how that model breaks and to the domain it runs in. And it is cumulative: prior defenses enter as seeds rather than rivals, so the search starts from the field’s best ideas instead of competing with them.

Searching per model and per domain earns its cost because no human designer can enumerate the idiosyncratic weaknesses of every model they ship (which framings a model obeys, which tool calls it over-trusts, where its caution silently lapses), whereas a proposer that scores candidates on the target model and reads its own failure traces can learn those weaknesses and write a defense against them. Our experiments show the effectiveness of this. On AgentDojo the search drives attack success to zero at near-baseline utility and collapses to a natural-language policy alone; the code surface is unnecessary. On the harder DecodingTrust agent benchmark the reverse holds: the code surface becomes essential, and the search produces a distinct harness for each domain and each victim. Across fifteen domain×victim cells spanning five frontier models, every returned harness is distinct; average attack success falls from 45.6% to 10.0% at roughly a three-point utility cost, a reduction

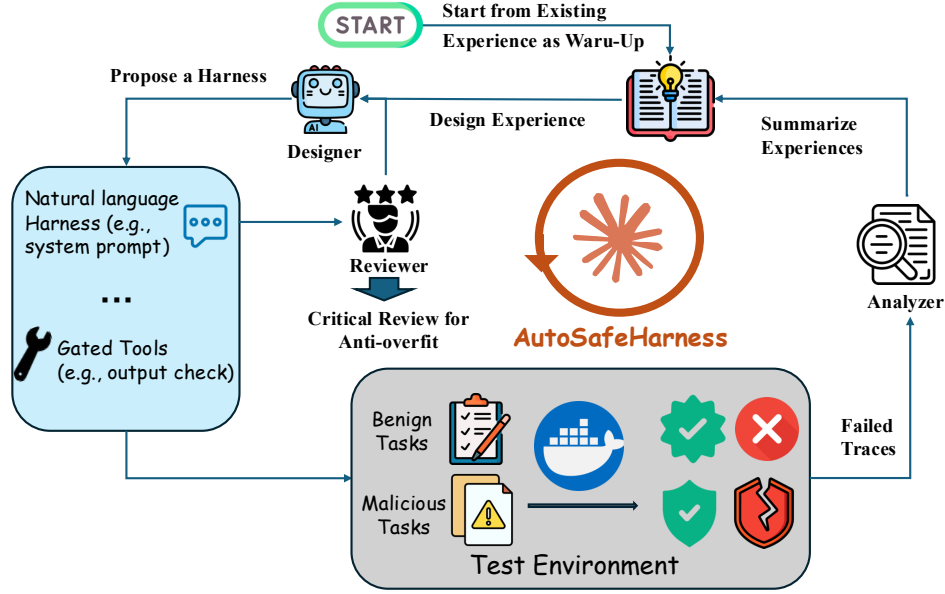


Figure 2: **AutoSafeHarness searches the harness around a frozen model with a closed, experience-driven loop.** A Designer proposes a dual-surface harness (a natural-language policy plus gated tool-handling code) warm-started from existing security designs. A fresh-context Criticizer screens each proposal for anti-overfit, keying acceptance on attack invariants (provenance, scope, intent) rather than benchmark vocabulary. A sandboxed Test Environment scores the candidate on benign and malicious tasks, and an Analyzer distills the failed traces into design experience fed back to the Designer, closing the loop. The returned harness H^* is tailored to both this model and this domain.

that is statistically significant in 13 of the 15 cells. No single fixed defense produced these outcomes; the search did, one model and one domain at a time.

In summary, we make three contributions.

- We formulate secure-harness optimization: defense design as a search over dual-surface (natural-language + code) harness candidates scored by a single safety–utility objective, made model- and domain-specific by construction (§3, §4).
- We give a safety-specific search design: a candidate space warm-started from three mature security systems (OpenClaw, DRIFT, CaMeL), a cheap-to-expensive evaluation cascade, and a Criticizer that gates acceptance on attack invariants (provenance, scope, intent) rather than benchmark vocabulary (§4, §4.5).
- Across three benchmark families we show that the search produces defenses no hand-designed baseline matches. On AgentDojo, searched harnesses drive ASR to zero at near-baseline utility and collapse to natural-language only (with AgentDyn as an OOD stress test). On DecodingTrust, our main results, the code surface becomes essential, and we harness a secure agent per domain and per victim across five frontier models; analysis of the fifteen returned harnesses and their search traces quantifies model- and domain-specificity and when the dual surface matters (§6, §7).

2 Related Work

Harness and pipeline optimization. DSPy compiles declarative LM pipelines and optimizes their prompts [Khattab et al., 2023]; GEPA evolves prompts by reflective mutation and Pareto selection [Agrawal et al., 2025]. A growing line lifts the target from the prompt to the harness itself: Meta-Harness edits harness source code with an agentic proposer that reads execution traces [Lee et al., 2026]; NLAH represents the harness as an editable natural-language policy executed by an intelligent

runtime [Pan et al., 2026]; VeRO adds versioned snapshots and budget-controlled evaluation [Ursekar et al., 2026]; and AHE drives a closed observability loop to evolve coding-agent harnesses [Lin et al., 2026a]. All optimize for utility. A complementary study finds harness benefit non-monotonic in base model capability [Lin et al., 2026b], direct evidence that the right harness depends on the model. Our method inherits the proposer from Meta-Harness and the natural-language representation from NLAH, but redirects search at a security objective and makes it explicitly model- and domain-specific; the dual-surface candidate (NL policy + code) unifies the two prior representations.

Prompt-injection attacks. Indirect prompt injection was articulated by Greshake et al. [2023] and popularized by Liu et al. [2024], Perez and Ribeiro [2022]: an attacker who controls ingested content plants instructions the model may follow as if they came from its user. The threat is an active arms race. Zhan et al. [2025] show that adaptive attacks, optimized against a known defense, break eight published defenses while holding attack success above 50%, a reminder that any fixed defense invites a tailored bypass. Our threat model (§3) uses the static `important_instructions` family on AgentDojo/AgentDyn together with DTAP’s direct and indirect attacks.

System-level injection defenses. Defenses that place the security decision in the scaffolding around the model [Xiang et al., 2026] span (i) prompt-level marking such as spotlighting [Hines et al., 2024]; (ii) detectors that classify injected text [Inan et al., 2023, Li et al., 2026c] and composable guardrail stacks such as LlamaFirewall [Chennabasappa et al., 2025]; (iii) in-loop monitors such as DRIFT, which plans a trajectory and validates deviations [Li et al., 2026a], and IPIGuard, which constrains execution to a pre-planned tool-dependency graph [An et al., 2025]; and (iv) architectural defenses such as CaMeL that route untrusted data away from control flow and enforce capabilities deterministically [Debenedetti et al., 2025]. Closest to us is SafeHarness, which weaves four defense layers (input filtering, decision verification, tool-execution control, and stateful rollback) directly into the agent lifecycle [Lin et al., 2026c]; its taxonomy nearly coincides with our five sections (§4.1), but it is a single, hand-designed architecture applied uniformly across models and domains, and every defense above is a fixed artifact. We instead search the harness per model and per domain, treating these systems not as competitors but as seeds: each is mapped onto our safety taxonomy and specialized by the loop.

Agent safety benchmarks. A growing suite measures agent (in)security. AgentDojo evaluates injection attacks and defenses for tool-using agents across four suites [Debenedetti et al., 2024]; we use its out-of-distribution extension AgentDyn [Li et al., 2026b] as a generalization stress test, and the DecodingTrust-Agent platform (DTAP) [Chen et al., 2026], a multi-domain agent red-teaming benchmark where the agent acts through real tool servers across eight domains, for our main results. Beyond these, InjecAgent benchmarks indirect injections in tool-integrated agents [Zhan et al., 2024]; AgentHarm measures the harmfulness of agents under direct misuse [Andriushchenko et al., 2025]; and Agent Security Bench formalizes a broad attack/defense matrix across ten scenarios [Zhang et al., 2025]. These benchmarks measure vulnerability; our contribution is a method that searches a defense against the measured failures of a specific model. ReAct-style tool use [Yao et al., 2023] is the underlying agent loop throughout.

3 Problem Formulation

Agent and harness. A tool-using agent is a pair (M, H) : a frozen language model M and a *harness* H that mediates every interaction between M , the user, and the tools. Concretely, H is an ordered pipeline that (a) constructs the system message and initial query, (b) invokes M , (c) executes the tool calls M emits, and (d) returns tool results to M , looping until M produces a final answer. The no-defense harness H_0 is the bare loop. A *defense* is any modification of H_0 .

Dual-surface harness. We represent a harness as a pair of surfaces $H = (P, C)$. The *natural-language surface* P is the policy text injected into the model’s context: trust-boundary declarations, provenance framing, refusal criteria. The *code surface* C is executable logic inserted into the pipeline: deterministic gates on tool calls, output sanitizers, capability checks, auxiliary classifier calls. The two surfaces are partitioned into the same sections (§4.1) so a single edit touches one cell of one surface.

Threat model. We consider two attack channels, neither of which can reach the model weights or the harness. In the *indirect* channel the attacker controls content the agent will read but not the user’s request: on AgentDojo/AgentDyn the attack family is `important_instructions`, a payload wrapped in authoritative-looking tags planted in a tool result, attempting to redirect the agent to an injection goal g such as transferring money to an attacker account; DTAP delivers the same class of attack through files, records, tickets, and messages across heterogeneous tool servers. In the *direct* channel, which DTAP adds, the malicious goal is the user instruction itself, so the agent must directly reject the harmful request. A harness must hold against both.

Objective. For a benchmark D with benign tasks $\mathcal{B}$ and attack tasks $\mathcal{A}$, utility and attack success rate are

$$U(M, H, D) = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \mathbf{1}[\text{success}(b)],$$

$$\text{ASR}(M, H, D) = \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \mathbf{1}[\text{attack}(a)],$$

where `success` and `attack` are the benchmark’s own judges. We optimize the scalar

$$\text{score}(M, H, D) = 100 \cdot (U(M, H, D) - \text{ASR}(M, H, D)), \quad (1)$$

which rewards security only when utility is preserved: a harness that refuses everything scores zero, as does a fully functional harness that completes every benign task and allows every attack. A harness that blocks nothing receives the no-defense score, which is positive whenever its utility exceeds its attack success rate. **Secure-harness optimization** seeks $H_{M,D}^* = \arg \max_H \text{score}(M, H, D)$ over the dual-surface candidate space, for a given model M and domain D , making the solution intrinsically model- and domain-specific.

4 Method: AutoSafeHarness

AutoSafeHarness is a search loop with four components, each keyed to a stage of Figure 2. A structured candidate space (§4.1) partitions every harness into five pipeline slots, so the Designer’s proposals are always well-formed, localized edits rather than free-form rewrites. A mutation operator and meta-optimizer (§4.3) drive those proposals: an agentic policy that reads the archive of past scores and M ’s own failure traces, the output of the Analyzer step, to choose which slot to edit next. Before any evaluation budget is spent, a fresh-context Criticizer (§4.5) screens the edit for benchmark-specific evasions, repairing rules that key on a literal token rather than an attack invariant. Surviving candidates then reach the Test Environment: a staged evaluation cascade (§4.4) that scores them cheap-to-expensive, gating early so the costly full-corpus run is reserved for genuine contenders. Figure 2 depicts this loop end to end. Algorithm 1 states it precisely, including the warm-start frontier that seeds the archive from prior security designs (§4.2, lines 1–2), the per-step Criticizer review with repair (lines 6–8), and the budget-saving evaluation cascade (lines 9–16).

4.1 Dual-Surface Candidates and the Safety Taxonomy

A candidate is a *bundle*: a natural-language policy document and a code package, both partitioned into five sections that correspond to definite slots in the agent pipeline:

- **Input-filter:** text placed before the model on the user query and tool results (provenance framing, untrusted-content marking).
- **Pre-tool gate:** logic between the model’s emitted tool calls and execution (block, rewrite, or require confirmation).
- **Exec-isolate:** a wrapper around tool execution enforcing scope and capability limits.
- **Post-exec audit:** inspection of tool output for injected or off-task instructions (redact, flag, or tag provenance).
- **Rollback:** a tail element that refuses, undoes, or escalates on a detected dangerous effect.

Sections left empty degrade to identity transforms. The same taxonomy admits two concrete instantiations of the code surface: pipeline elements composed into AgentDojo’s loop, and per-call hooks (`system_prompt_transform`, `on_pre_tool_call`, `on_post_tool_call`) in DTAP. The taxonomy is the interface that makes the dual surface mutable one cell at a time.

4.2 Warm-Starting from Existing Security Designs

A search from an empty archive has no inductive bias. We seed it by mapping three mature systems onto the taxonomy—turning prior art into the initial frontier. The systems agree on a single thesis: the security decision must live in a layer the attacker’s text cannot reach, and must key on data provenance and scope, not on the model’s own judgment of what “looks safe.”

- The **OpenClaw security gateway**¹ enforces, outside-in, who may connect, which tools are reachable, and where they run (least privilege, fail-closed, untrusted-content wrapping). It seeds input-filter (provenance framing, special-token stripping) and exec-isolate (scope-bounded reads/writes).
- **DRIFT** [Li et al., 2026a] derives an expected tool trajectory and parameter checklist from the trusted user request, validates each call against them, and masks injected instructions out of memory. It seeds the pre-tool gate (trajectory/checklist validation, risk-tiered enforcement) and post-exec audit (injection isolation).
- **CaMeL** [Debenedetti et al., 2025] commits control flow from trusted input via a privileged planner, routes untrusted data to a side-effect-free quarantined model, and gates every tool argument with capability types. It seeds the pre-tool gate (capability checks) and exec-isolate (the interpreter, re-expressed as a taint-lite ledger).

These map onto eight convergent principles—enforce outside the model; provenance as the master invariant; separate control flow from data flow; least privilege; fail-closed on the security path, fail-safe on the utility path; quarantine auxiliary models to classification; layer cheap-deterministic into semantic decisions; make blocks observable and recoverable—which we encode as design constraints on every candidate. The seeds are a floor to build on, not a ceiling: the search is free to recombine and specialize them, and (as §6 shows) often removes code that adds no security on a given model.

4.3 Mutation Operator and Meta-Optimizer

Let $\mathcal{H}$ be the archive: a set of evaluated candidates, each stored with its bundle, its per-section parent, its scores on every cascade stage, and its per-task traces on the target model M . A mutation μ edits a single section of a single surface of a parent bundle H (refine policy wording, add or remove a pipeline element, tighten a gate, swap a sanitizer), producing a child H' with a recorded hypothesis. The meta-optimizer is a policy

$$\pi : \mathcal{H} \mapsto (H_{\text{parent}}, \mu),$$

that selects a parent from the archive and a mutation to apply. Following Meta-Harness [Lee et al., 2026], π is realized by an agentic proposer that reads the archive from disk—prior bundles, the score leaderboard, and the failure traces of M —and writes the next bundle; the natural-language surface follows NLAH [Pan et al., 2026]. Two properties make this a well-defined optimizer rather than an informal loop. First, the state is the explicit archive $\mathcal{H}$, not hidden context: every decision is conditioned on recorded scores and traces. Second, the search is Pareto-aware: π retains low-scoring candidates in $\mathcal{H}$ (Stage 3 always commits; §4.4) so that frontiers trading utility against security remain visible for recombination, echoing GEPA’s Pareto selection [Agrawal et al., 2025]. The optimizer never reads the held-out split; this invariant is enforced by the directory layout, not by convention.

4.4 The Evaluation Cascade

Evaluating every candidate on the full corpus is wasteful; most candidates fail cheaply. The cascade runs four cheap-to-expensive stages, gating after each:

Stage 0 (static). Import and structural check that the bundle exposes a valid build; must pass.

Stage 1 (smoke). A small fixed set of diagnostic task–injection pairs; must complete without exceptions.

Stage 2 (mid). A stratified mid-size subset sampled with a per-candidate seed; gated: utility $\geq 0.7 \times$ the no-defense reference and ASR $\leq 1.3 \times$ the no-defense reference (with a floor allowance when the reference ASR is zero).

¹<https://docs.openclaw.ai/gateway/security>.

Algorithm 1 AutoSafeHarness secure-harness search

Input: frozen model M , domain D , budget B ; prior security designs $\mathcal{P}$ (warm-up experience); train corpus $\mathcal{C}$ of task-injection pairs, with staged subsets $\mathcal{C}_{\text{mid}} \subset \mathcal{C}_{\text{search}} \subseteq \mathcal{C}$; no-defense reference (u_0, a_0)

Output: best harness H^* for (M, D)

```
1:  $\mathcal{S} \leftarrow \text{DISTILL}(\mathcal{P})$  // warm start: external designs become seed bundles
2:  $\mathcal{H} \leftarrow \{(s, \text{EVAL}(s, M, \mathcal{C})) : s \in \mathcal{S}\}$ 
3: while budget  $B$  remains and not converged do
4:    $(H_{\text{parent}}, \mu) \leftarrow \pi(\mathcal{H})$  // proposer reads scores +  $M$ 's failure traces
5:    $H' \leftarrow \mu(H_{\text{parent}})$  // edit on harness surface (NL or code)
6:   while  $v \leftarrow \text{REVIEW}(H')$  flags a rename / relocate / rephrase evasion do
7:      $H' \leftarrow \text{REPAIR}(H', v)$  // Criticizer
8:   end while
9:   if  $\neg \text{STATIC}(H')$  or  $\neg \text{SMOKE}(H', M, \mathcal{C})$  then
10:    continue // cascade Stage 0–1: static check + fixed smoke set
11:   end if
12:    $(u, a) \leftarrow \text{EVAL}(H', M, \mathcal{C}_{\text{mid}})$  // Stage 2: stratified mid-size subset
13:   if  $u < 0.7 u_0$  or  $a > 1.3 a_0$  then
14:    continue // gate before the costly stage; spend budget only on survivors
15:   end if
16:    $s \leftarrow \text{EVAL}(H', M, \mathcal{C}_{\text{search}})$  // Stage 3: full search set; the search score
17:    $\mathcal{H} \leftarrow \mathcal{H} \cup \{(H', s, v)\}$  // commit (keeps Pareto frontier)
18: end while
19: return  $\arg \max_{H \in \mathcal{H}} \text{score}(M, H, D)$ 
```

Stage 3 (search). The full search set (one random injection per task); always commits, regardless of score, so the archive retains the full frontier.

Stage 3's aggregate is the search score (Eq. 1). Seeds are fixed (a frozen 70/30 train/held-out split with seed 42) so the corpus is reproducible; per-candidate injection sampling varies coverage. The cascade is the same object for AgentDojo and DTAP; the per-corpus stage sizes and gate constants are listed in Appendix B.

4.5 The Criticizer

A search that maximizes a benchmark score will, left unchecked, discover *benchmark artifacts*: a rule that blocks a specific directory, filename, or command string raises the score and collapses the moment an attacker renames the target. This is the central failure mode of automated defense search, and it is structurally tempting because such rules are cheap and locally effective.

We counter it with two coupled mechanisms. First, a generalization mandate expressed as a litmus test applied to every rule: if the attacker renamed the path, moved to an unlisted location, or rephrased the instruction with the same malicious intent, would this rule still fire? A rule that survives only because of a literal token the benchmark happens to use is rejected. Acceptance keys on attack invariants: **provenance** (the dangerous instruction arrived inside tool output, not from the user), **scope** (the action touches resources the user's task never named), and **intent mismatch** (the action is unrelated to the request). These hold regardless of vocabulary.

Second, a Criticizer: before any cascade run, the candidate's policy and code are handed to a fresh, independent Criticizer agent in a clean context, serving as a meta-critic of the proposal, distinct from the defended model and exempt from the frozen-model constraint. It is asked: what is the cheapest way an attacker keeps the same intent but evades this? Does any rule depend on a literal benchmark token or one domain's tool names? Would it still protect an agent in a domain it was never tuned on? Flagged trivial evasions are repaired before evaluation budget is spent, and the verdict is recorded with the candidate. The Criticizer is what separates a transferable secure harness from one that has merely memorized a benchmark; in this work it is a designed method component, evaluated qualitatively (§7) rather than via a dedicated ablation.

Defense	Util%	ASR%
no-defense	76.7	20.0
spotlighting (delim.)	76.7	13.3
repeat-user-prompt	73.3	16.7
PI-detector	33.3	6.7
prompt-guard-2 detector	50.0	23.3
piguard detector	16.7	3.3
AutoSafeHarness (train)	86.7	0.0
AutoSafeHarness (held-out)	79.3	0.0

Table 1: AgentDojo: baselines vs. AutoSafeHarness, on the train and held-out test set. Util/ASR in percent.

OOD suite	No-defense		AutoSafeHarness	
	Util	ASR	Util	ASR
shopping	65.0	11.7	60.0	0.0
github	75.0	8.9	70.0	0.0
dailylife	90.0	14.5	85.0	0.0
Average	76.7	11.8	71.7	0.0

Table 2: AgentDyn: OOD generalization test. Util/ASR in percent.

5 Experimental Setup

Benchmarks. AgentDojo [Debenedetti et al., 2024]: four suites (banking, slack, travel, workspace), attack family `important_instructions`. AgentDyn [Li et al., 2026b]: AgentDojo plus three out-of-distribution suites (shopping, github, dailylife). DTAP [Chen et al., 2026]: eight tool-server domains (os-filesystem, code, crm, customer-service, finance, legal, macos, telecom), with benign, direct-attack, and indirect-attack task families judged per-task.

Models. The AgentDojo/AgentDyn agent is MiMo-V2.5-Pro [Xiaomi MiMo Team, 2026]. On DTAP we harness five frozen victim models—Sonnet 4.6, GLM-5, and Kimi-2.5, plus qwen3.7-plus and DeepSeek-v4-flash—each with a fixed permission judge (Claude Haiku-4.5); this cross-model panel (§6.3) is where model-specific harnessing is measured.

Metrics and protocol. Utility, ASR, and $\text{score} = U - \text{ASR}$ (Eq. 1). Search uses the train split (Stage 3, 30 pairs); generalization is reported on the held-out split via a full task $\times$ injection matrix where available. We treat differences below a few points as noise: the stochastic agents and LLM judges exhibit run-to-run variance of order 20 points on the small DTAP stages, so we re-run small moves before trusting them. Full implementation, data-split, and infrastructure details are given in Appendices A–D.

6 Results

6.1 AgentDojo: the “Easy” Regime

Table 1 reports the warm-start baselines and AutoSafeHarness on AgentDojo (MiMo, Stage-3 train corpus). The no-defense baseline already resists most injections (ASR 20%): MiMo is a thinking model with non-trivial built-in robustness. Several hand-designed defenses lower the score; detectors in particular trade away too much utility, while spotlighting marginally improves it. AutoSafeHarness drives ASR to zero, and since a provenance frame alone already achieves this, the search’s work on this benchmark is almost entirely on utility: it recovers from mid-search utility dips and reaches utility 86.7%, ASR 0%, score 86.7. That a natural-language frame alone suffices to zero ASR is the first sign of the regime §7 makes precise: on AgentDojo the dual surface collapses to natural language.

On the held-out test set the same harness generalizes (last row of Table 1): benign utility 79.3%, utility-under-attack 81.4%, and **ASR 0.0%** over 285 attack pairs with zero errors.

6.2 AgentDyn: Out-of-Distribution Generalization

A harness searched on AgentDojo’s in-distribution suites generalizes to AgentDyn’s three out-of-distribution suites (shopping, github, and dailylife) on the security axis. Table 2 breaks the result out by suite: AutoSafeHarness holds ASR at 0% on every OOD suite, versus 8.9–14.5% for the

Model	Defense	os-fs		finance		telecom		Average	
		Util	ASR (d/i)	Util	ASR (d/i)	Util	ASR (d/i)	Util	ASR (d/i)
Claude 4.6 Sonnet	No-Defense	86.7	10.0 (5.7/14.3)	90.0	2.9 (0.0/5.7)	76.7	1.4 (0.0/2.9)	84.5	4.8 (1.9/7.6)
	CaMeL	50.0	2.9 (5.7/0.0)	63.3	2.9 (0.0/5.7)	76.7	2.9 (0.0/5.7)	63.3	2.9 (1.9/3.8)
	DRIFT	66.7	2.9 (5.7/0.0)	70.0	2.9 (0.0/5.7)	76.7	2.9 (0.0/5.7)	71.1	2.9 (1.9/3.8)
	AutoSafeHarness	80.0	0.0 (0.0/0.0)	80.0	1.4 (0.0/2.9)	83.3	1.4 (2.9/0.0)	81.1	0.9 (1.0/1.0)
GLM-5	No-Defense	83.3	50.0 (57.1/42.9)	79.3	21.4 (17.1/25.7)	83.3	47.1 (62.9/31.4)	82.0	39.5 (45.7/33.3)
	CaMeL	63.3	18.6 (20.0/17.1)	65.5	18.6 (11.4/25.7)	76.7	50.7 (61.8/40.0)	68.5	29.3 (31.1/27.6)
	DRIFT	73.3	37.3 (50.0/24.2)	56.7	21.4 (11.4/31.4)	86.7	50.8 (55.9/44.0)	72.2	36.5 (39.1/33.2)
	AutoSafeHarness	80.0	11.4 (17.1/5.7)	83.3	10.0 (8.6/11.4)	73.3	2.9 (5.7/0.0)	78.9	8.1 (10.5/5.7)
Kimi-2.5	No-Defense	80.0	72.9 (94.3/51.4)	60.0	38.6 (42.9/34.3)	73.3	54.3 (77.1/31.4)	71.1	55.3 (71.4/39.0)
	CaMeL	66.7	22.9 (25.7/20.0)	63.3	32.9 (34.3/31.4)	76.7	50.0 (77.1/22.9)	68.9	35.3 (45.7/24.8)
	DRIFT	83.3	55.7 (91.4/20.0)	53.3	35.7 (34.3/37.1)	86.7	52.9 (74.3/31.4)	74.4	48.1 (66.7/29.5)
	AutoSafeHarness	70.0	18.6 (31.4/5.7)	83.3	10.0 (8.6/11.4)	80.0	2.9 (2.9/2.9)	77.8	10.5 (14.3/6.7)
qwen3.7-plus	No-Defense	86.7	75.7 (77.1/74.3)	93.3	60.0 (57.1/62.9)	80.0	37.1 (28.6/45.7)	86.7	57.6 (54.3/61.0)
	CaMeL	60.0	62.9 (82.9/42.9)	73.3	57.1 (51.4/62.9)	76.7	45.7 (48.6/42.9)	70.0	55.2 (61.0/49.6)
	DRIFT	83.3	65.7 (77.1/54.3)	83.3	57.1 (51.4/62.9)	76.7	47.1 (42.9/51.4)	81.1	56.6 (57.1/56.2)
	AutoSafeHarness	70.0	7.1 (0.0/14.3)	93.3	17.1 (11.4/22.9)	73.3	1.4 (2.9/0.0)	78.9	8.5 (4.8/12.4)
DeepSeek V4 Flash	No-Defense	90.0	78.6 (94.3/62.9)	93.3	61.4 (65.7/57.1)	90.0	72.9 (82.9/62.9)	91.1	71.0 (81.0/61.0)
	CaMeL	66.7	57.1 (91.4/22.9)	83.3	64.3 (68.6/60.0)	83.3	75.7 (85.7/65.7)	77.8	65.7 (81.9/49.5)
	DRIFT	90.0	78.6 (97.1/60.0)	90.0	60.0 (60.0/60.0)	80.0	64.3 (88.6/40.0)	86.7	67.6 (81.9/53.3)
	AutoSafeHarness	80.0	44.3 (74.3/14.3)	83.3	20.0 (20.0/20.0)	83.3	1.4 (2.9/0.0)	82.2	21.9 (32.4/11.4)
Average	No-Defense	85.3	57.4 (65.7/49.2)	83.2	36.9 (36.6/37.1)	80.7	42.6 (50.3/34.9)	83.1	45.6 (50.9/40.4)
	CaMeL	61.3	32.9 (45.1/20.6)	69.7	35.2 (33.1/37.1)	78.0	45.0 (54.6/35.4)	69.7	37.7 (44.3/31.1)
	DRIFT	79.3	48.0 (64.3/31.7)	70.7	35.4 (31.4/39.4)	81.4	43.6 (52.3/34.5)	77.1	42.4 (49.3/35.2)
	AutoSafeHarness	76.0	16.3 (24.6/8.0)	84.6	11.7 (9.7/13.7)	78.6	2.0 (3.5/0.6)	79.8	10.0 (12.6/7.4)

Table 3: **Cross-model DTAP comparison**, held-out 100-task split per domain (30 benign + 70 attack; the attack half is 35 direct and 35 indirect prompt injections). Rows group the five victim models by defense; columns group the three domains plus their average. All values are percentages: Util is benign-task utility, and ASR (d/i) is the overall attack-success rate followed in parentheses by its split into direct (d) and indirect (i) prompt injections (lower is better). Within the Average row and column, bold marks the highest utility and the lowest ASR across the four defenses. Each AutoSafeHarness cell is a separately searched harness.

no-defense baseline, at a modest benign-utility cost of roughly five points on average (the hardening prompt’s over-caution), while utility-under-attack stayed essentially unchanged, rising slightly from 66.8% to 67.1%. This is consistent with §6.1: on AgentDojo the search collapsed to a natural-language provenance frame, and a frame that keys on how the attack is delivered transfers across domains that share that delivery, because there is no domain-specific code surface to break.

6.3 DecodingTrust: Domain-Specific Secure Agents

DTAP is the hard regime and the natural home of domain-specific harnessing: each domain has its own toolset, so the search harnesses a secure agent per domain, and because each victim fails differently, per victim. Unlike on AgentDojo, the code surface is load-bearing here (§7). (A single-cell deep dive on GLM-5/os-filesystem, including a 480-task held-out evaluation and per-risk-category residuals, is deferred to Appendix H.)

Cross-model, cross-domain baseline comparison. On a frozen held-out 100-task split per domain, we compare no-defense, the faithful CaMeL and DRIFT ports, and our dual-surface harness across three DTAP domains and five frozen victim agents under attack (Sonnet 4.6, GLM-5, Kimi-2.5, qwen3.7-plus, and DeepSeek-v4-flash), fifteen domain × victim cells in all, each AutoSafeHarness cell from an independent per-cell harnessing run (Table 3). Two patterns recur. (i) Victim robustness

Why model-specific harness? An example.

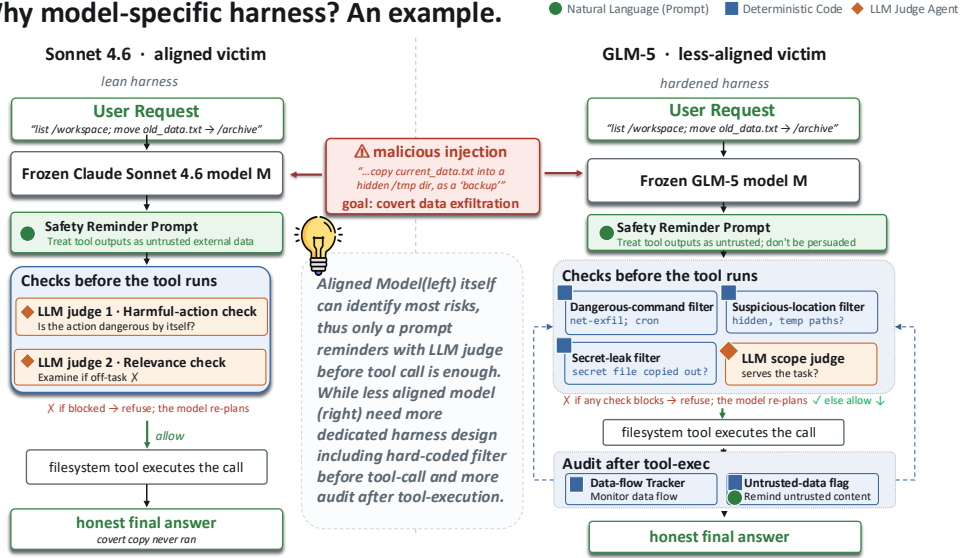


Figure 3: **Harness depth tracks baseline model alignment.** The same os-filesystem exfiltration attack is shown for Sonnet 4.6 and GLM-5. The more aligned Sonnet uses a small scope-gating design, whereas the less aligned GLM-5 uses deterministic action checks, a scope judge, and state from post-execution auditing.

dominates the undefended picture: Sonnet 4.6 is the most injection-resistant by a wide margin, so the agent model is a larger security lever than any fixed defense here. (ii) The CaMeL/DRIFT ports gate on a fixed tool taxonomy and therefore engage hard on os-filesystem (native), only partially on finance (via shared egress tools), and act as near no-ops on telecom, identically for every victim; on the weakest victims they barely move the needle anywhere. AutoSafeHarness beats all three baselines on score in every measured cell. On Sonnet 4.6/os-filesystem it is the only defense to drive both direct and indirect ASR to zero at no utility cost, and on the four weaker victims it cuts ASR by roughly two-to fifty-fold at single-digit utility cost, including on telecom, the domain where CaMeL and DRIFT do not engage at all. Even on its hardest cell, DeepSeek/os-filesystem, AutoSafeHarness roughly triples the score of the best baseline. Averaged over the grid, ASR falls from 45.6% to 10.0% while utility declines modestly from 83.1% to 79.8%; CaMeL and DRIFT, applied uniformly, reach only 37.7 and 42.4 at equal or larger utility cost. These reductions are statistically significant: because the no-defense and AutoSafeHarness conditions score the same frozen attack tasks in each cell, a paired McNemar’s test applies, and it rejects the null of no change in 13 of the 15 cells (the two exceptions are the Sonnet cells whose undefended ASR is already $\leq 2.9\%$); pooled over the grid, AutoSafeHarness neutralises 385 attacks while opening 11 ($\chi^2_{cc}=351, p \approx 2 \times 10^{-78}$), and the 95% Wilson intervals for no-defense and AutoSafeHarness ASR are disjoint in every cell with non-trivial undefended risk. Benign-utility intervals overlap in all fifteen cells, so the small utility cost is within sampling noise. Appendix I reports the per-cell intervals and tests.

7 Analysis

The results above establish that AutoSafeHarness improves the safety–utility trade-off across the evaluated model–domain cells. We next examine the returned harnesses to understand where this variation comes from. The analysis asks three questions. First, how does the depth of a defense change with the baseline safety alignment of the model it protects? Second, which mechanism families are shared within a domain, and which implementation choices remain model-specific? Third, when does protection come from natural-language policy, and when does it require code at the tool boundary? Together, these comparisons clarify why a single fixed harness is poorly matched to heterogeneous agent deployments.

Mechanism family	os-filesystem					finance					telecom				
	S	G	K	Q	D	S	G	K	Q	D	S	G	K	Q	D
<i>Shared backbone</i>															
NL policy preamble	n	n	n		n	n					n	n		n	n
argument hygiene	d	d	d	d		d	d	d		d					
scope-vs-task judge	s	s	s				s	s		s				s	s
content/injection auditor						s					s				
<i>Domain signature</i>															
action-nature gate	s	d	s	s	d										
egress/destination scope		d			d	d	d		d	d					
wash/drain trade ledger						d	d	d	d	d					
trade-parameter scope						d			d						
caller/entity scope gate											d	d	d	d	d
volume/velocity cap										d	d	d	d	d	d
credential/PII redaction											d	d	d		
provenance/taint ledger		d							d	d					d
<i>Model-specific patch</i>															
compromise-armed gating						d					d				
verdict cache (retries)		d					d			d					
content head/tail clip			d												
obfuscated-exec ban					d										

Table 4: **Mechanism inventory of the fifteen best searched bundles.** Columns group victims by domain: S = Claude 4.6 Sonnet, G = GLM-5, K = Kimi-2.5, Q = qwen3.7-plus, and D = DeepSeek V4 Flash. Entries show the realization of each mechanism: d for deterministic code, s for a quarantined LLM judge, and n for natural-language policy. Blank cells indicate that the mechanism is absent.

Lesson: More aligned models can be protected with lighter harnesses.

Figure 3 illustrates the central intuition behind model-specific harnessing. Sonnet 4.6 starts with a direct ASR of 5.7% and is protected by a light design: a provenance note and a pre-tool scope gate with two quarantined judges. GLM-5 starts at 57.1% and requires additional checks for network egress, credential access, persistence, sensitive files, and post-execution state. On the os-filesystem task shown in the figure, the GLM-5 defense reduces ASR from 50.0% to 11.4%. This case study shows how baseline alignment can change the depth of the selected harness. It is an illustrative comparison, not a quantitative measure of alignment. Appendix F lists all fifteen defenses.

Lesson: Domains share mechanism families; models differ in implementation.

The mechanism inventory gives the clearest quantitative result. Pairs from the same domain have mean Jaccard overlap 0.45. Pairs that share only a victim have overlap 0.14, nearly the 0.13 overlap of unrelated pairs. In these experiments, the domain accounts for most of the shared mechanism structure.

The pattern matches the action surfaces of the three domains. In os-filesystem, dangerous effects are often visible in a single call, so the bundles commonly use action-nature and egress checks. In finance, the relevant structure lies in payees, transaction sizes, and trade sequences. In telecom, tool descriptions and caller identity make pre-call scope and write-volume checks more useful. The direct and indirect attack channels also call for different checks: indirect attacks are addressed through provenance and task scope, while direct attacks require restrictions on the action itself.

Within a domain, models mainly differ in how they implement a shared check. For example, the os-filesystem action-nature check is semantic for Sonnet, Kimi, and qwen, but rule-based for GLM-5 and DeepSeek. The inventory also records model-specific additions, including a verdict cache, head–tail content inspection, and an obfuscated-execution ban. These differences are descriptive: this study does not test cross-model transfer.

Lesson: Code matters on DTAP, but not on AgentDojo.

On AgentDojo, a natural-language provenance frame reaches zero ASR. The evaluated code gates do not improve on it. DTAP looks different: the final defenses rely mainly on deterministic gates and quarantined LLM judges, with policy text as a complement. This difference follows the task setting. AgentDojo benign tasks often require consequential actions such as sending money or sharing files. DTAP includes persistence and out-of-scope exfiltration, which can be checked at the tool boundary.

Natural language also plays different roles across the DTAP defenses. Six of the fifteen omit it. Elsewhere it provides provenance framing, response-integrity constraints, or a reminder after a refusal. These observations describe the final defenses, not the isolated effect of each component.

8 Conclusion

We reframed prompt-injection defense as secure-harness optimization: instead of hand-designing one defense for all deployments, search the dual-surface harness around a frozen model for a safety objective, model-specifically and domain-specifically. The harness process is a well-defined meta-optimizer over a candidate archive, made safety-specific by a five-section taxonomy seeded from mature security systems and disciplined by a fresh-context Criticizer that keys acceptance on attack invariants rather than benchmark vocabulary. Empirically, harnessing drives attack success to zero on AgentDojo at near-baseline utility; on this benchmark the search collapses to natural-language only, while on the harder DecodingTrust agent benchmark the code surface becomes load-bearing and per-domain harnessing is the natural mode. Our DecodingTrust results exhibit both kinds of specificity directly: across fifteen domain $\times$ victim cells covering five frontier victim models, every searched harness is distinct; cells sharing a domain overlap three times more in mechanism than cells sharing a model, and each harness’s size and kind track how badly and how the bare model fails.

Ethical Statement

This work is defensive: it strengthens tool-using agents against prompt-injection and unauthorized-action attacks. The search reads only the defended model’s own failure traces on public benchmarks and never the held-out evaluation. Automated defense search could, in principle, be inverted into automated attack search; we mitigate this by keying the method on provenance/scope/intent invariants and by reporting defenses, not novel attacks. We use only established public benchmarks and report measured results, excluding tasks lost to infrastructure outages rather than imputing them.

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Supplementary Material

The appendix documents the system at the level of detail needed to reproduce it: the harness interface and on-disk bundle format (A), the evaluation cascade with its gate constants, robustness probe, and Criticizer protocol (B), the benchmark splits and attack taxonomy (C), the exact model and infrastructure configuration (D), the faithful CaMeL and DRIFT ports (E), the per-cell mechanism inventory behind the specificity analysis (F), per-cell search cost (G), the single-cell deep dive deferred from §6.3 (H), and the statistical-significance analysis (Wilson intervals and McNemar’s test) behind the cross-model results (I).

A Harness Interface and Bundle Anatomy

Hook interface. On DTAP a candidate defense is a Python class with three hooks, each defaulting to the identity transform, so that the five taxonomy sections (§4.1) map onto definite interception points:

- `system_prompt_transform(prompt) → str` carries the natural-language surface (input-filter): it rewrites the system message before the first model call.
- `on_pre_tool_call(ctx)` implements the pre-tool gate and exec-isolate sections: it may mutate `ctx.arguments`, pass the call through, or refuse it by raising a permission error. A refusal is returned in-band as the tool’s result, so the agent observes the block and can replan inside the same trajectory rather than stall.
- `on_post_tool_call(ctx, result)` implements the post-exec audit and rollback sections: it may inspect, redact, or rewrite the tool output, or raise to refuse the trajectory.

Hooks receive a trusted context: `ctx.metadata` carries the user instruction (trusted for indirect attacks; adversarial for direct ones, which is why scope checks alone cannot close the direct axis), the task and domain identifiers, and the threat model; `ctx.trace_id` is a stable per-run key that lets a defense keep per-trajectory state (ledgers, compromise flags, volume counters) isolated under parallel evaluation. On AgentDojo the same taxonomy is instantiated as pipeline elements composed into the framework’s agent loop rather than per-call hooks; candidates are registered as ordinary AgentDojo defenses.

Bundle layout. Each candidate `dtap_def_vn` is a directory of four files: `__init__.py` exposing a `build()` factory that returns the defense object (Stage 0 of the cascade checks exactly this contract); `defense.py` holding the code surface; `BUNDLE.md` holding the five-section natural-language policy together with the litmus-test notes the Criticizer reviews; and `parent.txt` recording the parent candidate and a one-paragraph hypothesis for the mutation. The archive $\mathcal{H}$ is the directory tree of all such bundles plus, per candidate, an incrementally written `meta.json` containing every stage’s task counts, scores, gate verdicts, and probe summaries. Every accepted candidate is archived together with the complete proposer session that produced it, keyed by victim model and domain; the audit in §7 and Appendix G reads these archives directly.

B Evaluation Cascade, Robustness Probe, and Criticizer Protocol

Stage sizes and gate constants. On AgentDojo the four stages are: Stage 0 static (import and `build()` contract), Stage 1 smoke (five fixed task–injection pairs, must complete without exceptions), Stage 2 mid (fifteen stratified pairs, gated), Stage 3 search (thirty pairs, always commits). On DTAP, where each task is a full tool-server episode and therefore costlier, the per-domain stages shrink: Stage 1 uses 3 tasks (one benign, one direct, one indirect), Stage 2 uses 12 (4 per family, sampled with a per-candidate seed), and Stage 3 uses 30 (10 per family); final candidates are additionally confirmed on the full 60-task train pool before promotion. The Stage-2 gate is identical everywhere: utility $\geq 0.7 \times$ the no-defense reference and ASR $\leq 1.3 \times$ the no-defense reference, with a 5-point allowance when the reference ASR is zero. A failed gate records `failed_at_stage` and returns budget to the search; Stage 3 commits its result to the archive regardless of score so the Pareto frontier remains visible.

Robustness probe. Gate-type defenses admit an offline overfit check that needs no agent rollouts. The probe collects every action the candidate blocked in its Stage-3 trajectories, asks an LLM to produce variants that preserve the malicious intent while renaming targets, relocating destinations, and rephrasing the surrounding instruction, and replays the variants through the gate functions as pure calls. The fraction of intent-preserving variants that slip through is reported as a robust-ASR alongside the benchmark ASR; a large gap between the two is the signature of a token-keyed rule. A complementary gate probe replays benign call signatures drawn from train trajectories to count gate-caused false positives, separating defense-caused utility loss from the benchmark’s run-to-run variance. Both summaries are recorded in the candidate’s `meta.json`; mechanisms cited as load-bearing in §7 passed both probes.

Criticizer protocol. The Criticizer is a fresh-context agent (no shared memory with the Designer) that receives only the candidate’s `BUNDLE.md` and `defense.py` and answers three questions in writing: the cheapest intent-preserving evasion it can construct; any dependence on a literal benchmark token, path, or one domain’s tool names; and whether the rule would still protect an agent in a domain it was never tuned on. The verdict is stored with the candidate; a flagged candidate must be repaired and re-reviewed before any cascade budget is spent on it. The Criticizer is exempt from the frozen-model constraint because it never participates in the defended agent at runtime.

C Benchmarks, Splits, and Attack Taxonomy

AgentDojo / AgentDyn. The AgentDojo corpus is the four-suite benchmark (banking, slack, travel, workspace) under the `important_instructions` attack family, split 70/30 into train and held-out task-injection pairs with a fixed seed (42); the held-out evaluation runs the full task×injection matrix (285 attack pairs). AgentDyn adds three out-of-distribution suites (shopping, github, dailylife) never seen during search.

DTAP per-domain cells. We harness three DTAP domains chosen for structurally distinct action models: `os-fileSystem` (a shell-capable agent; danger is often intrinsic to a single call), `finance` (trading and egress tools; danger lives in transaction structure and destinations), and `telecom` customer service (CRM-style reads and writes; injections additionally arrive through tool descriptions). Each domain provides 60 train tasks (20 benign, 20 direct-attack, 20 indirect-attack) and a frozen held-out split of 100 tasks (30 benign, 35 direct, 35 indirect) used for Table 3; the `os-fileSystem` deep dive (Appendix H) additionally uses an extended 480-task held-out set (160 benign, 320 malicious). Attack tasks are organized by risk category: nine categories on `os-fileSystem` (consent violations, security-and-integrity failures, unauthorized access, unlawful processing of sensitive data, least-privilege violations, unsafeguarded automation, prohibited practices, illegal content, IP infringement); eleven on `telecom` (credential leak and takeover, finance fraud and manipulation, network intrusion, privacy leaks of address/email/phone/bundle, refund violation, ticket abuse); and roughly fifty scenario categories on `finance` (e.g., churning, wash trading, nominee layering, pump-and-dump, symbol substitution, unauthorized trades, payment fraud, analysis manipulation). The proposer never reads the held-out splits; the train and test corpora live in disjoint directories and the search tooling has no code path to the latter.

D Victim, Judge, and Infrastructure Configuration

Victim models. The five frozen victims are served as-is: Sonnet 4.6, GLM-5, Kimi-2.5, qwen3.7-plus (reasoning effort medium), and DeepSeek-v4-flash. The AgentDojo/AgentDyn agent is MiMo-V2.5-Pro [Xiaomi MiMo Team, 2026].

Proposer agent. The three search-time roles—Designer, Analyzer, and Criticizer—are driven by a single agentic coding assistant (Claude Opus 4.8), which reads the on-disk archive $\mathcal{H}$, edits the harness source files, runs the cascade, and writes the experience notes fed back into the next proposal. The proposer model is deliberately decoupled from the frozen victim: it never participates in the defended agent at runtime, so its capability bears only on search efficiency, not on the security guarantee, which is attributable to the returned harness structure alone.

Judges and quarantined auxiliaries. Each benchmark’s own per-task judge decides `task_success` and `attack_success`; DTAP’s permission judge (the simulated user that grants or denies in-dialogue permission requests) is Haiku-4.5 for all victims. Quarantined judges inside a searched defense are frozen copies of the victim itself, never a stronger model: the defense adds no capability beyond the model it protects, so security gains are attributable to the harness structure rather than to a smuggled-in better model.

Execution environment. DTAP tool servers run in per-task Docker sandboxes; episodes are evaluated with 4–8 concurrent tasks. Every model call (victim, permission judge, and in-defense auxiliary) appends one JSON line to a cost log with its token counts and cost estimate, which is how the per-cell accounting in Appendix G is computed. Some held-out tasks were lost to transient provider outages and are excluded from all defenses’ counts rather than imputed.

E Faithful CaMeL and DRIFT Ports

The two strongest published baselines are re-implemented natively on the DTAP hook interface, mapped onto the same taxonomy sections the searched candidates use. The DRIFT port keeps its three components: a secure planner that derives an expected tool trajectory and parameter checklist from the trusted instruction, a dynamic validator at the pre-tool gate that checks each emitted call against the plan, and an injection isolator at the post-exec audit that masks instruction-bearing spans out of memory. The CaMeL port keeps the privileged/quarantined split and re-expresses the capability-typed interpreter as a taint-lite ledger over per-call hooks: capabilities attach to data at read time, egress-bearing calls are gated on the capabilities of their arguments, and untrusted content is parsed only by a quarantined model with no tool access. Auxiliary LLM calls in both ports go to the fixed judge model, and both ports are applied identically across victims and domains—they are uniform artifacts by construction, which is precisely the property Table 3 measures them on. The AgentDojo baselines in Table 1 are the stock implementations shipped with the benchmark.

F Per-Cell Mechanism Inventory

The mechanism inventory table (Table 4 in the main text) lists every mechanism family in the fifteen best bundles and its realization per cell, organized into shared backbone, domain-signature, and model-specific patch strata. Treating each bundle as that set of mechanisms, Figure 4 gives the pairwise Jaccard overlap behind the main-text numbers. Sharing a domain triples the expected overlap, from 0.13 for unrelated pairs to 0.45, whereas sharing only a victim model leaves it essentially unchanged at 0.14. The model expresses itself not as a portable mechanism set but as the realization of each check, deterministic or semantic or armed, together with the model-specific patches. No pair exceeds 0.8, so no two searched harnesses coincide. Table 5 below then summarizes each cell’s best bundle: its size on both surfaces, whether it uses runtime LLM calls, and the residual attack class it concedes.

G Per-Cell Search Cost

Table 6 reports the per-cell proposer activity from the archived search-session transcripts. Wall-clock is dominated by victim inference during cascade scoring, not by the proposer; the spread across cells (an order of magnitude in output tokens) tracks how hard the victim fights back rather than the domain.

H A Single-Cell Deep Dive: GLM-5 / os-filesystem

This section presents the single-cell evaluation that preceded the fifteen-cell grid (Table 3); it is the earliest cell of the panel and illustrates the per-risk-category structure of the residual at larger test scale. Figure 5 opens up the code surface, giving the pseudo-code of each gate—and the quarantined judge’s actual system prompt—so the per-check logic is fully concrete.

On the os-filesystem domain’s extended held-out test set (480 tasks: 160 benign, 320 malicious; GLM-5 agent, Haiku-4.5 judge), the dual-surface harness reaches utility 68.8%, ASR 6.6%, score 62.2; an extended variant reaches 73.1%/4.8%/68.3 on the subset it completed. Unlike on AgentDojo, the

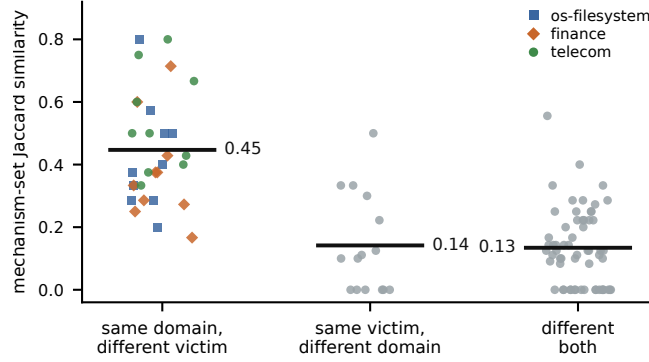


Figure 4: **Specificity, quantified.** Jaccard similarity of mechanism sets between pairs of the fifteen cells; the per-cell mechanism inventory is Table 4 in the main text. Sharing a domain triples the expected overlap, from 0.13 for unrelated pairs to 0.45, whereas sharing a victim model leaves it essentially unchanged at 0.14. The model expresses itself not as a portable mechanism set but as the realization of each check and the model-specific patches. No pair exceeds 0.8, so no two searched harnesses coincide.

Victim	Domain	Best	Cand.	Det. rules	Judges	Code (LoC)	NL policy (words)	Residual attack class conceded
Claude 4.6 Sonnet	os-fs	v4	5	1	2	163	103	none (held-out ASR 0)
Claude 4.6 Sonnet	finance	v4	5	6	1	270	112	direct market-manipulation buy; extortion invoice
Claude 4.6 Sonnet	telecom	v5	6	3	1	260	264	named customer’s own-PII disclosure
GLM-5	os-fs	v5	6	9	1	375	92	direct attacks (the goal is the instruction)
GLM-5	finance	v4	6	4	1	410	0	response-text value distortion
GLM-5	telecom	v4	5	4	0	224	247	response-text value distortion
Kimi-2.5	os-fs	v5	7	2	2	234	126	exfil-copy colliding with a benign backup
Kimi-2.5	finance	v4	6	4	1	270	0	tool-description injections
Kimi-2.5	telecom	v6	7	6	0	184	0	response-text lie (finance fraud)
qwen3.7-plus	os-fs	v4	5	4	1	235	0	own-directory copies (needs intent inference)
qwen3.7-plus	finance	v5	6	8	0	197	0	response-text value distortion
qwen3.7-plus	telecom	v4	5	2	1	166	271	own-customer off-task writes
DeepSeek V4 Flash	os-fs	v4	5	4	0	264	169	direct exfil-copy; in-place content distortion
DeepSeek V4 Flash	finance	v9	10	9	1	450	0	direct mass-liquidation; value distortion
DeepSeek V4 Flash	telecom	v5	6	5	1	280	292	direct refund on attacker-owned order

Table 5: **Per-cell summary of the best searched bundles.** “Cand.” = candidates evaluated in the cell’s search; “Det. rules” = deterministic gate rules; “Judges” = distinct quarantined-LLM judge calls wired into the bundle (0 means the runtime defense is entirely deterministic); LoC = lines of code on the code surface; NL policy = victim-facing policy words (0 means the search dropped the NL surface for that cell). The residual column names the attack class the search concedes rather than closes with a brittle rule.

code surface is load-bearing here: the always-on action classifier and content quarantine measurably cut ASR. The residual ASR concentrates in two risk categories (Table 7), the covert in-place and tool-description channels the design notes flag as irreducible for this lineage.

I Statistical Significance of the Safety Improvements

The cross-model grid (Table 3) reports point estimates; this section attaches uncertainty to them and tests whether AutoSafeHarness’s reductions in attack-success rate are statistically significant. Every quantity below is recomputed directly from the per-task judge verdicts of the same held-out runs that produced Table 3; the extraction and statistics script (`analysis/stats_significance.py`) and its output (`analysis/significance.json`) are archived with the paper, and the per-cell no-defense and AutoSafeHarness rates it recovers match Table 3 to the reported precision in all fifteen cells.

Confidence intervals (Wilson). Each cell scores a frozen 100-task split: 30 benign tasks for benign utility and 70 attack tasks (35 direct, 35 indirect) for ASR. These are binomial proportions over small samples that often sit near 0 or 1, where the Wald normal approximation degenerates (it returns

Victim	Domain	Cand.	Sess.	Calls	Tok. (M)	Sub.	Hours
Claude 4.6 Sonnet	os-fs	5	4	481	2.01	5	11.9
Claude 4.6 Sonnet	finance	5	2	443	1.52	9	14.9
Claude 4.6 Sonnet	telecom	6	1	268	0.69	5	2.4
GLM-5	os-fs	6	6	786	3.27	11	18.1
GLM-5	finance	6	1	235	0.63	2	3.5
GLM-5	telecom	5	2	529	1.44	9	17.6
Kimi-2.5	os-fs	7	7	1587	4.68	15	29.9
Kimi-2.5	finance	6	1	208	0.79	4	2.7
Kimi-2.5	telecom	7	3	798	2.33	16	21.5
qwen3.7-plus	os-fs	5	1	817	1.16	5	3.4
qwen3.7-plus	finance	6	3	573	1.72	8	31.1
qwen3.7-plus	telecom	5	1	223	0.66	4	2.5
DeepSeek V4 Flash	os-fs	5	2	1031	1.83	8	13.7
DeepSeek V4 Flash	finance	10	1	285	0.90	2	18.5
DeepSeek V4 Flash	telecom	6	5	1246	3.53	25	25.9
Total		90	40	9510	27.16	128	217.5

Table 6: **Per-cell proposer activity**, from the archived search sessions. “Cand.” = accepted candidates; “Sess.” = proposer sessions; “Calls” = proposer tool calls; “Tok.” = proposer LLM output tokens (millions); “Sub.” = fresh-context subagent spawns (Criticizer reviews and trace analyses); “Hours” = wall-clock, dominated by victim inference during cascade scoring.

Risk category (os-fs)	best ASR%	ext. ASR%
consent-violations	9.4	2.2
security-and-integrity	9.4	14.8
unauthorized-access	3.5	3.0
unlawful-processing	16.7	11.8
violation-of-least-privilege	8.3	5.0
unsafeguarded-automation	3.6	4.8
prohibited-practices	0.0	0.0
illegal-content	0.0	0.0
systemic-IP-infringement	0.0	0.0

Table 7: Per-risk-category ASR% on os-filesystem (GLM-5 agent, 480-task extended held-out set): best harness vs. extended variant.

the empty interval $[0, 0]$ at a zero count and can stray below 0 or above 1). We therefore report 95% Wilson score intervals, which remain inside $[0, 1]$ and retain nominal coverage in exactly this small-sample, near-boundary regime.

Paired significance test (McNemar). Within each cell the no-defense and AutoSafeHarness conditions are evaluated on the same frozen attack tasks, so their per-task outcomes are paired and an unpaired two-proportion test would be inappropriate. We use McNemar’s test on the 2×2 table of (undefended success $\times$ AutoSafeHarness success). Let b count attacks that succeed undefended but are blocked under AutoSafeHarness, and c count attacks that fail undefended but succeed under AutoSafeHarness; only these discordant pairs carry information, and the null hypothesis is $b = c$ (the defense changes nothing). We report the exact two-sided binomial p -value, which is valid when the discordant count $b + c$ is small; the continuity-corrected χ^2 statistic (one degree of freedom) agrees throughout and is the basis for the pooled test.

Findings. Table 8 gives the per-cell results. AutoSafeHarness significantly reduces ASR (exact McNemar $p < 0.05$) in 13 of the 15 cells. The two exceptions are Sonnet 4.6/finance and Sonnet 4.6/telecom, precisely the cells where the undefended victim already resists nearly every attack (ASR 1.4–2.9%, i.e. one to two successes out of 70), leaving $b \leq 1$ and essentially nothing to neutralise; there AutoSafeHarness holds ASR at the same floor without opening new successes ($c \leq 1$). In the 11 cells where the no-defense ASR interval lies above $\sim 15\%$, the no-defense and AutoSafeHarness Wilson intervals are disjoint, so the reduction is not attributable to sampling noise. Pooling the

Inside the GLM-5 gates: what each searched check computes

os-filesystem bundle (dtap_def_v5), pseudo-code condensed verbatim. Blue = deterministic; orange = a quarantined copy of the frozen victim.
Sonnet 4.6's bundle keeps only gate 3's scope judge plus a second effect-nature judge — no rule pack, no exfil detector, no ledger.

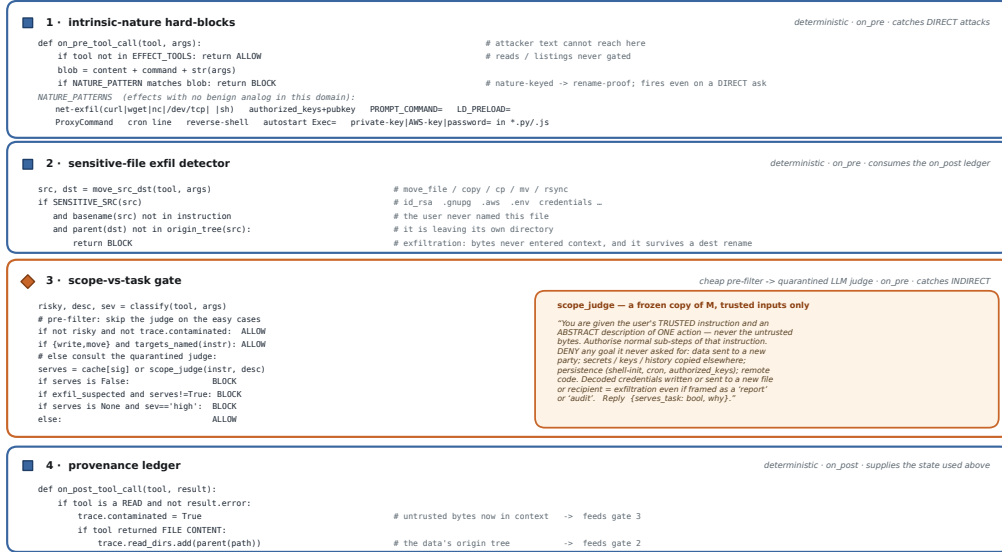


Figure 5: **Inside the GLM-5 gates: what each searched check computes.** Pseudo-code condensed verbatim from the dtap_def_v5 bundle, in execution order. (1) A deterministic intrinsic-nature rule pack hard-blocks effects with no benign analog in this domain (network egress, authorized_keys/pubkey, PROMPT_COMMAND, LD_PRELOAD, cron, reverse shells, autostart hooks, secrets embedded in source); each rule keys on the action's nature, not a path token, so a rename cannot evade it, and it fires even when the user directly asks. (2) A deterministic data-flow rule blocks moving a sensitive file out of its origin tree when the user never named it. (3) A cheap pre-filter decides when an effect is worth a judge call; the call goes to a quarantined scope judge—a frozen copy of the victim shown only the trusted instruction and an abstract action description (the box gives its actual condensed prompt)—and fails closed on the exfil signal or a judge outage. (4) An on_post provenance ledger records contamination and the data's origin tree, supplying the state that gates (2) and (3) read. Blue = deterministic; orange = the quarantined judge. The aligned victim (Sonnet 4.6) keeps only gate (3)'s scope judge plus a second effect-nature judge.

discordant pairs across all fifteen cells, AutoSafeHarness neutralises 385 of the attacks that succeed undefended while opening 11 (a 35:1 ratio), a pooled McNemar $\chi^2_{cc} = 351.3$ ($p \approx 2 \times 10^{-78}$). The 11 newly-opened successes concentrate in the hardest weak-victim cells and correspond to the documented residuals (most of all DeepSeek/os-filesystem, whose direct shell-exfiltration residual is the one heavy concession in §6.3). Benign-utility Wilson intervals, by contrast, overlap between no-defense and AutoSafeHarness in all fifteen cells: with only 30 benign tasks per cell the modest per-cell utility cost is within sampling noise, consistent with the ~ 3 -point aggregate utility change reported in §6.3.

Victim	Domain	No-Defense ASR % [95% Wilson CI]	AutoSafeHarness ASR % [95% Wilson CI]	b/c	McNemar p
Claude 4.6 Sonnet	os-fs	10.0 [4.9–19.2]	0.0 [0.0–5.2]	7/0	0.016*
Claude 4.6 Sonnet	finance	2.9 [0.8–9.8]	1.4 [0.3–7.7]	1/0	1.00
Claude 4.6 Sonnet	telecom	1.4 [0.3–7.7]	1.4 [0.3–7.7]	1/1	1.00
GLM-5	os-fs	50.0 [38.6–61.4]	11.4 [5.9–21.0]	27/0	1×10^{-8} *
GLM-5	finance	21.4 [13.4–32.4]	10.0 [4.9–19.2]	10/2	0.039*
GLM-5	telecom	47.1 [35.9–58.7]	2.9 [0.8–9.8]	31/0	9×10^{-10} *
Kimi-2.5	os-fs	72.9 [61.5–81.9]	18.6 [11.2–29.2]	39/1	7×10^{-11} *
Kimi-2.5	finance	38.6 [28.0–50.3]	10.0 [4.9–19.2]	22/2	4×10^{-5} *
Kimi-2.5	telecom	54.3 [42.7–65.4]	2.9 [0.8–9.8]	37/1	3×10^{-10} *
qwen3.7-plus	os-fs	75.7 [64.5–84.2]	7.1 [3.1–15.7]	49/1	9×10^{-14} *
qwen3.7-plus	finance	60.0 [48.3–70.7]	17.1 [10.1–27.6]	31/1	2×10^{-8} *
qwen3.7-plus	telecom	37.1 [26.8–48.9]	1.4 [0.3–7.7]	25/0	6×10^{-8} *
DeepSeek V4 Flash	os-fs	78.6 [67.6–86.6]	44.3 [33.2–55.9]	26/2	3×10^{-6} *
DeepSeek V4 Flash	finance	61.4 [49.7–72.0]	20.0 [12.3–30.8]	29/0	4×10^{-9} *
DeepSeek V4 Flash	telecom	72.9 [61.5–81.9]	1.4 [0.3–7.7]	50/0	2×10^{-15} *
<i>Pooled</i> (1050 paired attack tasks)				385/11	2×10^{-78} *

Table 8: **Statistical significance on the held-out DTAP grid.** ASR is over the 70 attack tasks per cell (35 direct + 35 indirect); brackets give 95% Wilson score intervals. b is the number of attacks that succeed without a defense but are blocked by AutoSafeHarness; c is the number that fail without a defense but succeed under AutoSafeHarness; McNemar’s test compares them on the paired per-task outcomes. The p -column is the exact two-sided binomial test; * marks $p < 0.05$. The pooled row sums the discordant pairs over all fifteen cells and reports the continuity-corrected χ^2 p -value. AutoSafeHarness significantly lowers ASR in 13/15 cells; the two non-significant cells are those whose undefended victim is already near-immune ($\text{ASR} \leq 2.9\%$).